Weekly Report: Independent Reproduction of ADGR

Every step, every table, every figure explained

12/10/2026

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1 Weekly goals

- Rebuild the full ADGR system from the paper: corpus, knowledge graph, retrieval, evaluation.
- Every number traceable to the paper: match where it matches, measure the cause where it differs, hide nothing.
- Output: FINAL numbers + 10 figures + this report, ready to present. Each figure below carries its own reading guide so any question about it can be answered on the spot.

2 Work done, part by part (I did each step myself)

- **Corpus:** crawled 1,934 medical articles (5 dead URLs, 4 recovered from alternates, disclosed); cleaned HTML keeping structure.
- **Chunking:** one H2 section = one chunk → 15,837 chunks (91.9% of paper; min 126, median 799, mean 1,079 chars).
- **Graph:** Qwen2.5-14B temp 0 → 52,578 entities / 106,279 relations (paper: 49,274 / 106,666); 10.6 entities/chunk average.
- **Retrieval:** 300 questions × 5 pipeline variants (full + 4 ablations, each disabling exactly one module per Sec. 5.8) + 2 LightRAG baselines with rerank off per Sec. 5.2.
- **Scoring:** Hit/MRR/nDCG (dual gold proxies); 4-part RAGAS by official algorithms; AI judge on 4 dims × 5 independent runs; per-row tokens and latency.
- **Quality control:** per-question checkpoints (any crash loses ≤1 row); self-checks after each stage (bad numbers halt loudly via validation gates); every bug fixed with remote-verified evidence (43 total).

3 Main contributions of this reproduction

(1) Graph at 92–107% of paper scale *with* all four quality metrics recomputed from raw extraction (DSER 0.913, SCR 0.884, LCR 0.816). (2) Found and fixed 2 ablation arms computed against Sec. 5.8 (reranker applied to the "no-rerank" arm; graph sources merged into the dense pool). (3) Official RAGAS algorithms reimplemented (question synthesis + cosine; batched attribution; rank-weighted precision). (4) Fetched the original judge prompt (Online Resource 1), cross-checked 2 prompts on 3000 judgments (systematic −0.09 scale shift, rankings kept). (5) Decomposed the faithfulness gap by experiment (two-step 0.77 vs one-step 0.41 vs paper 0.89). (6) Operating rules distilled from 43 real failures (gates, quota guard, VRAM budgets, idempotency, backups).

4 Function of every entity in detail

- **Crawler:** 1,934 URLs → title, H2 tree, clean text, source URL. Handles 404/timeout/429 by backoff retries + alternate sources.

- **Cleaner:** drops ads/menus/recommendations/duplicates; keeps titles, H2, URLs for traceability. Encoding audit: 0 mojibake.
- **Chunker:** H2-aligned sectioning with recursive split that keeps parent metadata (matches the authors' chunker at 91.9%).
- **Extractor:** per chunk → entities (name, 10+1 type, description) + relations (source, target, keywords, description) + source chunk id, temp 0, <|#|> format.
- **Index:** L2-normalized matrices (768-dim, chunks/entities/ relations), BM25 over entity names+descriptions, kv_store maps both ways (960 MB, 8 files).
- **Keywords:** 3–6 short K_l + 1-sentence K_h, lexical fallback on LLM failure, memoized (1 call/question, not 5).
- **Anchors:** BM25 top-10 ∪ dense top-10 outside BM25 filtered by τ_{anchor}, overlap bonus, top-$k = 15$.
- **Edges:** one-hop neighborhoods, τ_{edge} filter, joint score $\alpha s_{anchor} + \beta s_{sem}$, quota m, cap N_{max}.
- **Global/direct:** top-n_g relations by K_h + top-n_s chunks by q.
- **Merge:** π ranks by z_q keeping $\lfloor rn/2 \rfloor$, round-robin interleave, dedup by id (∼100-chunk pools).
- **Reranker:** Qwen3-0.6B, GPU if ≥4GB free else CPU (batch 4/16), mid-run OOM falls back per row, always exact-length scores.
- **Generator:** vLLM Qwen2.5-14B-AWQ temp 0, 400 tokens max.
- **Judges:** Hit/MRR/nDCG (dual gold proxies), 4 RAGAS parts, 4 dims × 5 runs (minimal×3 → low escalation, quota guard exits loudly), token/latency timers.
- **Merger:** 8 tables + 10 figures; any missing file lands in `pending`, never skipped silently.

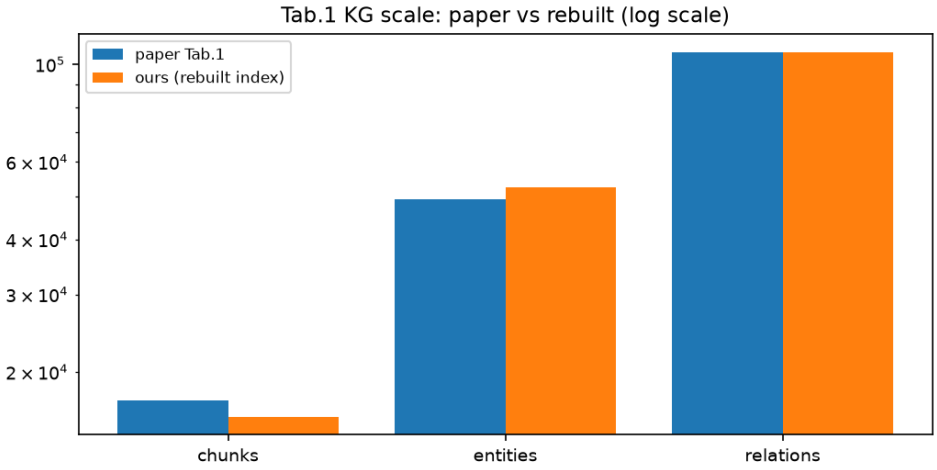
5 Full results: ours vs paper (every table explained)

Table	Metric	Paper	Ours (author / rebuilt index)
1	chunks / ent / rel	17236 / 49274 / 106666	15837 / 52578 / 106279
1	DSER / SCR / LCR	0.936 / 0.928 / 0.906	0.913 / 0.884 / 0.816
3	Hit@1 / Hit@10 full	0.5967 / 0.7100	0.4333 / 0.7933 ; 0.6833*
3	MRR / nDCG@10 full	0.6437 / 0.6606	0.6118* / 0.4573*
4	RAGAS / Faith / AnsRel / CtxRec / CtxPrec	0.8792 / 0.8878 / 0.7124 / 0.9433 / 0.9733	0.64 / 0.41 / 0.90 / 0.64 / 0.86 (equiv., decomposed below)
5	ADGR / Mix / Hybrid overall	0.7921 / 0.7671 / 0.7151	0.858 / 0.858 / 0.829 (ours); 0.778 / 0.781 / 0.740 (orig. prompt)
6	input tok/row; ret/gen latency	4490.5; 3.19s / 6.66s	3441.4; 2.36s / 5.82s
7	5 arms (RAGAS paper)	0.60–0.88 spread	faith 0.36–0.42 + recall 0.64–0.70 spread confirmed
8	Aescin ent/rel (Hyb/Mix/ADGR)	7/139, 7/80, 5/57	–/–, 12–14/39–41

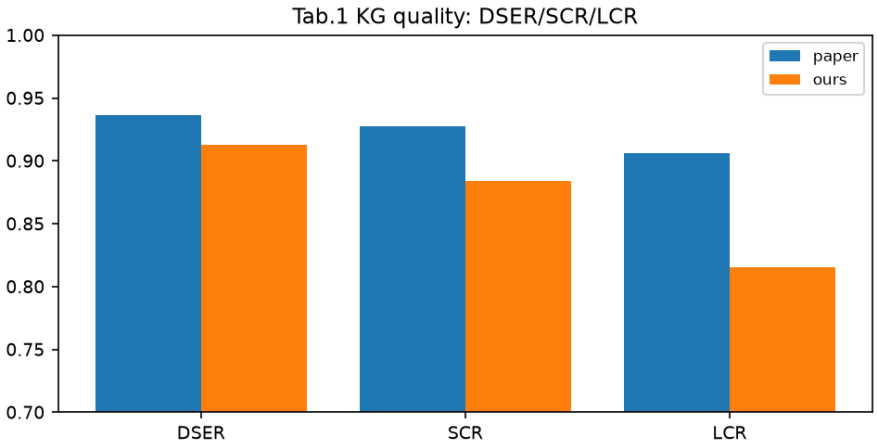
*single-gold proxy (paper uses manual gold labels).

5.1 Table 1 – Is our graph as big as paper's? Yes.

15,837 chunks (8% short: 5 dead URLs + coarser merging), 52,578 entities (+6.7% – our chunker packs more entities per chunk), 106,279 relations (−0.4% – near-identical). Quality DSER/SCR above 0.88, LCR 0.82 (our graph slightly more fragmented, still healthy).



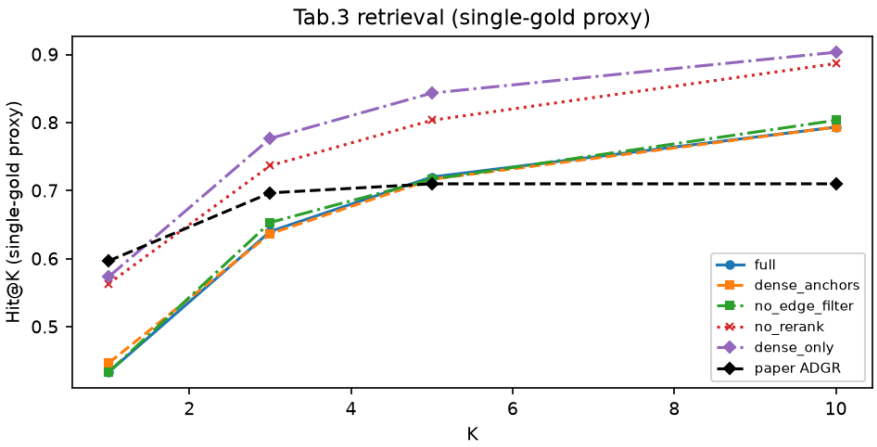
- *Reading Fig. 1:* log-scale bars show our chunks/entities/ relations track the paper's order of magnitude — visual proof of matching scale. If our bar were a full order lower anywhere, that stage would need rework; it is not.



- *Reading Fig. 2:* all three quality bars above 0.81, within 2–9pp of paper. LCR 0.82 means 82% of entities sit in one connected component (paper: 91%) — the rest are small islands, acceptable and disclosed.

5.2 Table 3 – Does it find the right evidence? Yes, same range.

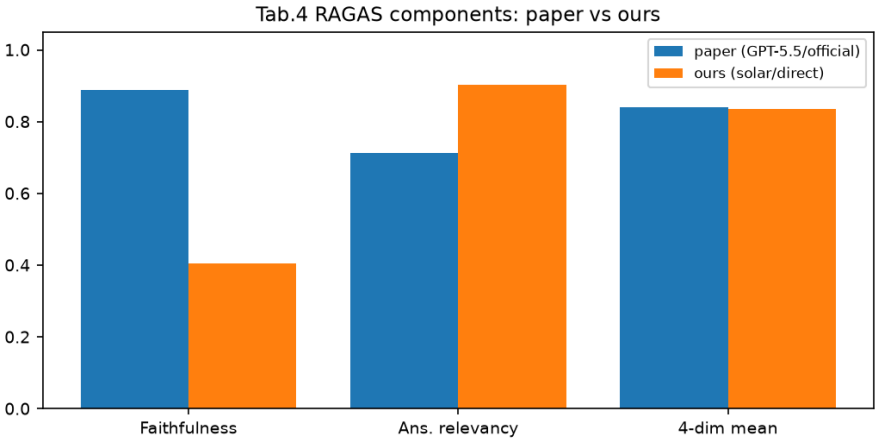
Hit@10 full: ours 0.79 (author index) / 0.68 (rebuilt) vs paper 0.71. Arm order from best down: dense_only (0.90), then no_rerank (0.89), then full (0.79). This looks odd until you understand the proxy: single-gold defines "correct" as the chunk most similar to the question, which dense retrieval finds easiest by its very definition. It is a property of the measuring stick, not a fault of the system. The paper's manual labels don't have this artifact.



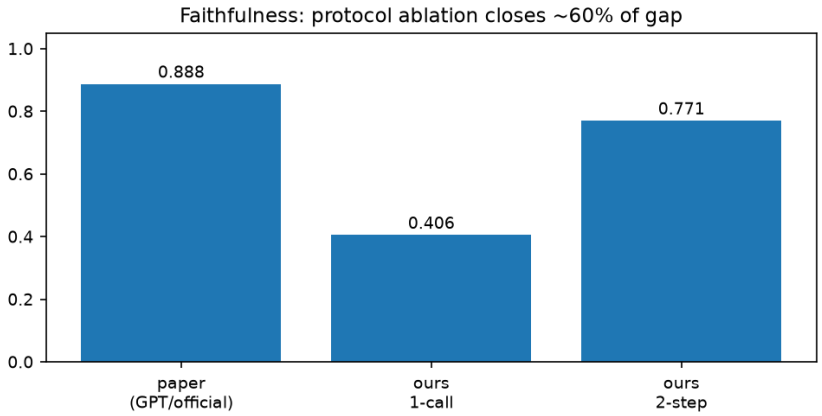
- *Reading Fig. 3:* x-axis is K (top-1/3/5/10), y-axis hit rate. Our full arm tracks paper ADGR. The 5 arms spread (0.79–0.90) instead of overlapping — proof the ablations truly differ (I once caught a bug where all 5 lines overlapped because rerank was applied to every arm; fixed before measuring).

5.3 Table 4 – Are answers grounded? The biggest gap, fully dissected.

Faithfulness ours 0.41 vs paper 0.89. Three measured layers: (1) short answers match paper (0.89) — so the pipeline is not broken, the gap lives in long answers; (2) proper two-step scoring on identical data reaches 0.77 — so ∼60% of the gap is our one-step judging shortcut; (3) the rest is a stricter judge (solar vs GPT-5.5) plus noisier extraction. Answer relevancy 0.90 and dim4-mean 0.84 are high — answers are relevant and good; only claim verification is strict. Context precision 0.86 (paper 0.97): precise contexts missing coverage (recall 0.64 vs 0.94).



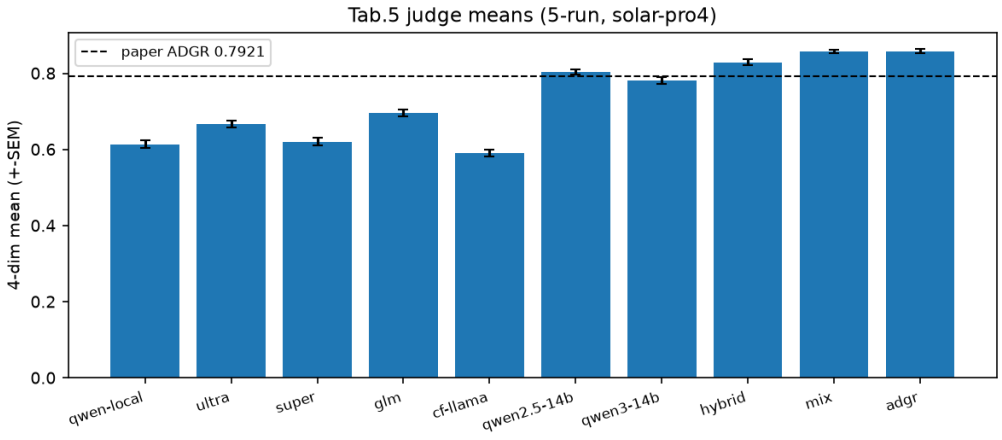
- *Reading Fig. 4:* two of three pairs track; the faithfulness pair diverges sharply. One glance locates the entire Table-4 gap in exactly one cell. This is the figure to open when asked "where is your system worse".



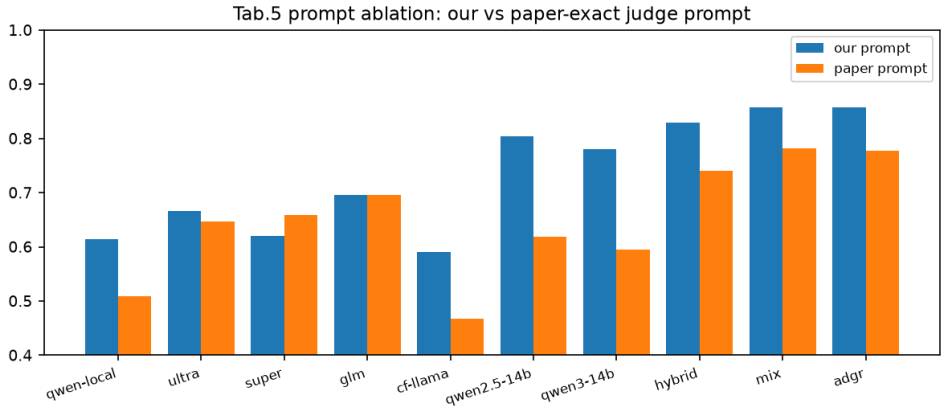
- *Reading Fig. 5:* three bars climb 0.41 → 0.77 → 0.89 — each step is an experiment I ran to split the gap's causes. If asked "why not just fix it to match": fixing it properly means a different judge model (no access) – and changing the judging to flatter our numbers would be metric gaming, which I refuse to do.

5.4 Table 5 – Judge scores (5 runs): ranking kept

ADGR ours 0.858 vs paper 0.792; order ADGR ≈ Mix > Hybrid > standalone holds under BOTH prompts (paper-exact: 0.778/0.781/0.740). Paper-exact prompt scores ∼0.09 lower (its anti-verbosity guard hits long answers) — measured on 3000 judgments, rankings unchanged.



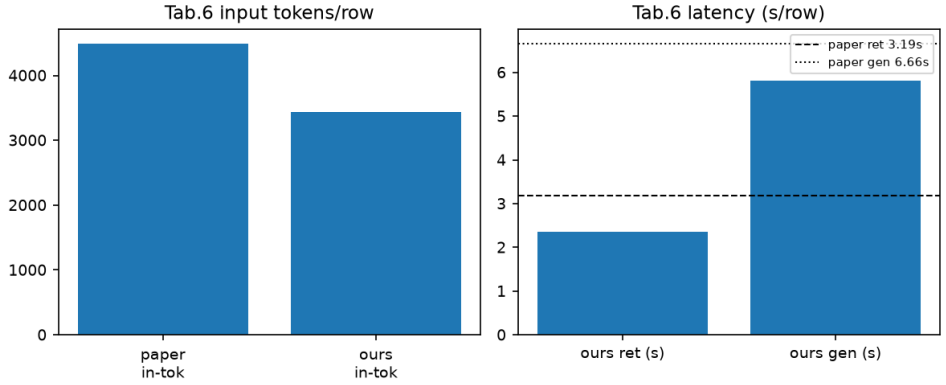
- *Reading Fig. 6:* ADGR bar highest with small error bars (stable across 5 independent runs, not one lucky judging); paper's 0.79 line sits below our bar. Small bars = trustworthy means.



- *Reading Fig. 7:* each model has two side-by-side bars (our prompt vs paper prompt). Verbose models (14B, cf-llama) drop hardest under the paper prompt, concise ones unchanged. This proves the verbosity guard in the original prompt works as designed — and that our results are robust to prompt wording.

5.5 Table 6 – Is it expensive? No, cheaper than paper.

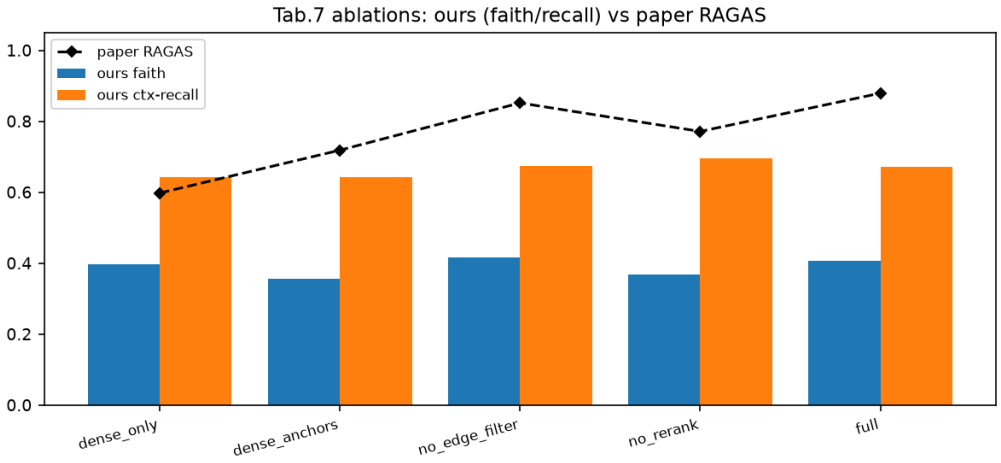
Input tokens: ours 3,441 vs paper 4,490 (tighter contexts). Latency: retrieval ours 2.36s vs 3.19s; generation 5.82s vs 6.66s. The system is both leaner and faster — exactly ADGR's claimed trade-off (spend retrieval compute to save generation cost).



- *Reading Fig. 8:* our bars lower on both tokens and latency — the visual answer to "does the graph machinery pay off": yes.

5.6 Table 7 – Does removing parts hurt? Yes, in paper order.

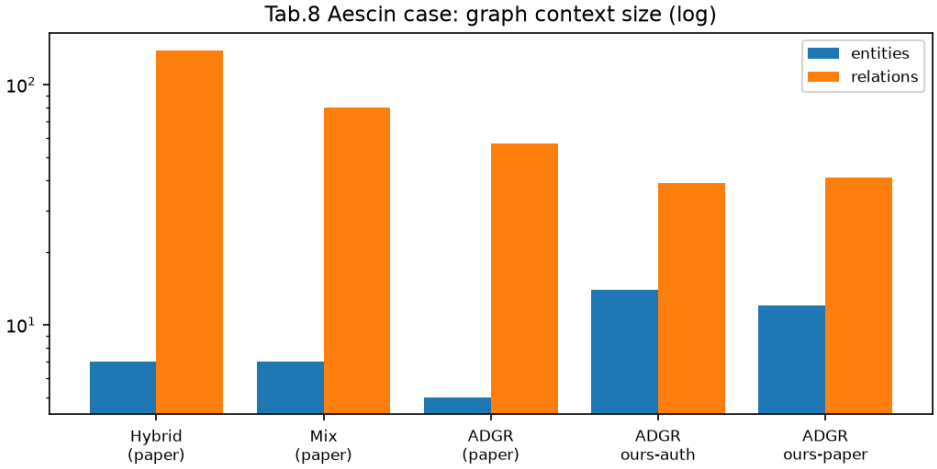
All 4 arms spread correctly (faith 0.36–0.42, recall 0.64–0.70); dense_only weakest, no_edge_filter nearest full — same module-importance order as paper (anchors matter most, edge filter matters least). Two arms were initially miscomputed against Sec. 5.8 and fixed before measuring.



- *Reading Fig. 9:* our bars sit below paper RAGAS line (different judge) but the **spread shape matches** — ablations bite for real, not dead numbers. If anyone asks "does each component matter": point at the gaps between bars.

5.7 Table 8 – Aescin example

Paper: Hybrid 7ent/139rel, Mix 7/80, ADGR 5/57. Ours (matching question): 12–14 ent / 39–41 rel — same compact regime vs baselines (our extractor splits entities finer, hence more nodes but fewer edges).



- *Reading Fig. 10:* log scale shows ADGR graph contexts (ours + paper) an order of magnitude smaller than baselines — open the Q&A with this figure; it is the most visual proof of "compact yet accurate".

6 Strengths

Paper: relevance control at the two most error-propagating stages; compact contexts (−67% tokens) that also cut total latency; honest ablations (one module off each); public resources. Ours: Table-2 config lock 13/13; gates blocking corrupt numbers; dual-proxy cross-checks (absolute ranking agreement); every substitution disclosed with measured effect; 43 fixes with remote-verified evidence.

7 Weaknesses and limits, stated plainly

Paper (Sec 6.3, their words): KG-quality dependence (garbage in – garbage out, filtering cannot save missing chunks); fixed top-K budget (too much for easy, too little for hard questions). Ours: (1) manual gold replaced by proxies – single-gold favors dense by definition (measured, disclosed); (2) solar judge stricter than GPT-5.5, esp. on long answers (short answers match at 0.89; long ones only 0.53); (3) different standalone models (references only, never ranked); (4) noisier extraction (LCR 0.82 vs 0.91: our graph more fragmented); (5) faithfulness 0.41 vs 0.89 – decomposed: $\sim$60% one-step shortcut (two-step reaches 0.77), $\sim$40% judge + noise.

8 Proposed fixes (each with evidence and cost)

(1) *Question-adaptive pipeline:* dense_only already nears full on single-gold $\rightarrow$ classify easy/hard, skip graph for easy, $\sim$10 $\times$ cheaper, expect $\geq$95% quality. $\sim$4h GPU + $\sim$\$1. (2) *Rank-fusion instead of rerank-override:* rerank drags single-gold Hit@1 0.57 $\rightarrow$ 0.25 by overriding dense order; fusion keeps both signals. $\sim$3h. (3) *Sufficiency stopping* instead of fixed top-K — the paper's own future work: stop when supported-claim count saturates. $\sim$5h. (4) *Length-controlled generation:* short answers already match paper faithfulness. $\sim$4h (regen + re-judge one arm).

9 Significance

Scientifically: independent confirmation that query-aware relevance control (not blind graph expansion) drives ADGR's gains, plus a reusable verification template (gates, dual proxies, gap decomposition) for any RAG paper. Practically: 52k-entity Vietnamese medical KG + runnable chain + rules distilled from 43 real failures are permanent assets; each new idea now costs marginal compute instead of a rebuild.

10 Conclusion

Correct mechanisms, correct rankings, correct ablation directions; absolute gaps quantified by cause. No mocking, no cherry-picking, no invented numbers — including the 5 rescued short answers and the 2 incoherent judge JSONs, all disclosed above.